

Cryptnox Smart Card Readers Manual

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Compatibility guides

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1 Smart Card Reader Compatibility by Country

Last reviewed: 27 August 2026.

This page answers one question, country by country: **will a Cryptnox USB smart card reader read my national ID or professional card, and what software do I also need?**

Two rules govern every row on this page.

1. The reader is necessary but not sufficient. Every national card scheme listed here also requires its own government middleware. A reader that meets the published specification will still do nothing until that software is installed. Where a scheme provides no freely installable software, this page says so.

2. Contact and contactless are different cards, not different modes. A card with gold pads on the surface is a *contact* card and needs the contact reader. A card read by holding it near the device is *contactless* and needs the contactless reader. Some cards have both interfaces; many have only one. Getting this wrong is the single most common cause of a reader "not working".

Countries on this page

- *Reader specifications*
- *Click-to-Tap on the contact reader*
- *How to read the verdicts*
- *Europe*
- *Outside Europe*
- *Travel documents — ICAO 9303*
- *Limitations*
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1.1 Reader specifications

	Contact reader	Contactless reader
Interface	ISO/IEC 7816, Class A/B/C (5 V / 3 V / 1.8 V)	ISO/IEC 14443 Type A **and** Type B, 13.56 MHz
Protocols	T=0 and T=1	106 / 212 / 424 / 848 kbps
Extended APDU	Yes	Yes
Stack	PC/SC 2.0, USB-IF CCID, Microsoft WHQL	PC/SC, USB 2.0 CCID
Card power	up to 500 mA	not applicable
Operating systems	Windows, macOS, Linux — no driver installation	Windows, macOS, Linux, Android via USB OTG
Connector	USB-C, with a USB-A adapter supplied	USB-C, with a USB-A adapter supplied

Extended APDU support is the specification that decides most of this page. Certificates and signatures on modern national cards routinely exceed the 255/256-byte short-APDU limit; readers without extended APDU fail precisely there, and the failure looks like a broken card rather than a broken reader. Both Cryptnox readers support it.

1.2 Click-to-Tap on the contact reader

The contact reader carries a button the contactless reader does not: **Click-to-Tap**. It solves one specific problem, and it has no bearing on any national-card verdict below.

FIDO2 and WebAuthn require a *user-presence* check on every authentication — proof that a person, and not software running in the background, approved the sign-in. A USB security key has a metal touch button for exactly this. A smart card sitting in an ordinary contact reader has nothing to touch, so the only gesture left is to pull the card out of the slot and push it back in, once per sign-in.

Click-to-Tap replaces that gesture. One press drives the smart-card interface through a controlled extraction-and-reinsertion sequence at the protocol level, with no mechanical movement of the card, so the card stays in the slot.

Requirement	Detail
Cards	Cryptnox FIDO2 smart cards only. With any other card the device behaves as a standard CCID / PC-SC contact reader.
Operating system	Windows. On macOS and Linux the button has no effect; the reader itself works normally on all three.
National eID cards	Not applicable. Click-to-Tap is a FIDO2 feature and does not change whether a government card can be read — that is decided by the country tables below.

The button is patented technology: registered Austrian utility model **GM55114/2025**, registered German utility model **No. 202025108028.1**, and United States patent application **No. 19/534.472** (pending).

Full detail, usage steps and troubleshooting: *Click-to-Tap: FIDO2 User Presence on a Desktop Reader*.

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1.3 How to read the verdicts

Verdict	Meaning
Works	Confirmed from the scheme's own published reader requirements *and* from users reporting generic PC/SC readers in service.
Likely	The interface and specification match, but community confirmation is thin or the scheme itself warns that some readers fail. Test before deploying at scale.
No	Interface mismatch, a mandatory homologation we do not hold, or the middleware does not support this class of reader.

1.4 Europe

1.4.1 Belgium — Belgian eID

Works with the contact reader. The Belgian eID is a contact card. Belgian users report that generic budget readers of any age drive it, and readers well over a decade old remain in service. The card also carries a contactless interface, but the desktop software path is the contact one.

Middleware: the free, open-source eID Middleware and eID Viewer.

Also covers the Kids-ID and the foreigner's card.

1.4.2 Czechia — eObčanka

Works with the contact reader. The Czech Ministry of the Interior publishes a standards-based specification rather than an approved-model list: ISO 7816, CCID, PC/SC, with Microsoft WHQL and plug-and-play operation recommended for Windows. The Cryptnox contact reader meets all of them.

Middleware: eObčanka klient — the card manager and identification tools.

1.4.3 Estonia — ID-kaart

Works with the contact reader. Estonia's ID-card runs on the standard USB CCID driver. The national ID authority notes that any reader capable of retrieving data from the card can be used, while cautioning that it has only tested certain models itself.

Middleware: DigiDoc4 and the ID-software suite.

1.4.4 Germany — Fahrerkarte and Personalausweis

These are two different cards with two different readers, and confusing them is the most common German support case.

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Card	Contact	NFC	Middleware	Verdict
Fahrerkarte (digital tachograph driver card)	Yes	No	tachograph/DVR software, varies by vendor	**Works** — contact reader
Personalausweis (nPA, eID function)	No	Yes	AusweisApp	**Works** — contactless reader
elektronische Gesundheitskarte (eGK)	Yes	partial	gematik TI stack	**No** — a gematik-approved terminal is mandatory

The Personalausweis eID function requires **extended length** APDU support: its certificates exceed 400 bytes and many inexpensive NFC readers cap at 261. AusweisApp states that a reader may work even if it is not on the tested list.

1.4.5 Hungary, Latvia, Romania, Slovakia

All four issue contact eID cards that work with the contact reader:

- Hungary — eSzemélyi · eSzemélyi Kliens
- Latvia — eID · eParaksts
- Romania — CEI · CEI app
- Slovakia — eID karta · eID klient

1.4.6 Italy — TS-CNS and CIE

TS-CNS works with the contact reader. The health/services card carries a contact chip and is driven by a per-card driver plus a smart card manager from the official list.

CIE 3.0 is contactless only — it has no contact pads. Italian sources caution that not every contactless reader drives it reliably, so this is a *Likely*, not a *Works*.

Middleware: CIE Software.

1.4.7 Luxembourg — eID

Contactless reader only. The Luxembourg government's own guidance specifies a contactless reader for the national eID.

Middleware: LuxTrust Middleware.

1.4.8 Netherlands — UZI-pas and ZORG-ID

Contact reader, with an important limit. The Cryptnox contact reader meets the UZI register's published reader specification — PC/SC, extended APDU support, and more than 60 mA of card power.

> **⚠ Warning**
>
> **UZI passes using 4096-bit RSA keys are a separate case.** SafeSign, the UZI middleware, keys its maximum APDU size to a list of reader names and applies a 255/256-byte default to any reader not on that list. A 4096-bit RSA signature is 512 bytes and cannot

be returned under that default. Cryptnox readers are **not currently on that list**, so a UZI pass carrying a 4096-bit key will not complete a signature until they are added. This is a software list, not a hardware limit — the reader itself supports extended APDU. Check the UZI register's approved-reader list and your own software's supported-reader list before ordering.

Middleware: SafeSign.

The Netherlands has no citizen contact card — DigiD is a phone and app scheme. The Belgian eID is widely used on Dutch desktops and is covered above.

1.4.9 Poland — e-dowód

Contactless reader. The Polish e-dowód's electronic layer is contactless; there is no contact chip to read. The reader meets the MSWiA technical requirements: ISO 14443 Type A and B, extended APDU, PC/SC.

Middleware: the free e-dowód software or the eDO App.

Poland's tachograph driver card is a contact card and **works with the contact reader**.

1.4.10 Portugal — Cartão de Cidadão

Works with the contact reader. Generic readers drive the card; note that some public service applications are reported to be fussy — that is the application, not the reader. The contactless interface is restricted to authorities.

Middleware: Autenticação.gov.

1.4.11 Spain — DNIe

Works with the contact reader.

Not with a desktop contactless reader. The DNIe chip is dual-interface, but the desktop software path is not: OpenSC's DNIe driver recognises the contact ATRs only and does not work over a contactless reader, and the official NFC route (DNIeRemote / DNIeSmartConnect) uses a *mobile phone* as the reader rather than a USB one.

Middleware: DNIe software.

1.4.12 Sweden — SITHS

Contact reader, employer-supplied software. On the hardware side the requirement is simply PC/SC compatibility, which the contact reader meets.

Note

SITHS middleware is **not a free public download**. Net iD requires a commercial licence, and Inera replaced it with SITHS eID in 2024. In practice the software reaches the user through their employer. Do not plan on installing it independently.

Finland's FINeID is a contact card and works with the contact reader, using mPollux DigiSign from DVV — note that DVV has stated this client is available only until the end of 2026.

1.4.13 Switzerland — health insurance card, SwissPass and the e-ID

Switzerland produces more wrong expectations than any other country on this page, because most of what a Swiss resident carries is **not** a PKI smart card.

Card	Contact	NFC	Verdict
Versichertenkarte / VeKa (health insurance card)	Yes	No	**Works** — contact reader; professional software
SwissPass (SBB)	No	Yes	Contactless reader, UID only — no certificate to authenticate with
Identity card (Identitätskarte)	No chip	No chip	**No reader applies** — see the 2026 note below
SuisseID	withdrawn	—	Discontinued; successor SwissID is an online account, not a card
e-ID (swiyu wallet)	—	—	Phone app; no card, and no NFC planned at launch

The **Versichertenkarte** is the one genuine Swiss contact smart card. Its chip follows eCH-0064 and ISO/IEC 7816-4, and it carries insurance data plus optional emergency data. Note the market: it is read by **medical practice software**, not by a free public download, so this is a professional use case rather than a consumer one.

SwissPass is a transport credential with ISO/IEC 14443 RFID tags. A contactless reader sees its UID; there is no certificate on it, so it cannot be used to sign or log in.

The Swiss identity card has no chip at all today. A biometric identity card is planned for **2 November 2026**, and it will carry a chip of the same family as the biometric passport — ICAO 9303, **contactless** — while a chip-free card remains on offer. Until then no reader, of either kind, does anything with a Swiss ID card.

The federal e-ID is not a card. It lives in the state swiyu wallet app, its introduction has slipped to around the end of 2026, and NFC is not part of the launch design. No reader is involved.

1.4.14 United Kingdom and Ireland

Neither country issues a national eID card. The **digital tachograph driver card** is a contact card and **works with the contact reader**, as does the **NHS Smartcard** in the UK, which is a contact smart card issued to health service staff and driven through the employer's identity agent software.

1.4.15 Countries where a desktop reader is the wrong tool

Published here because knowing what *not* to buy saves more time than another compatible row:

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Country	Why a USB reader is not the route
Austria	The Bürgerkarte was decommissioned in December 2023. ID Austria replaced it and authenticates by phone or FIDO2 token — no card reader is involved.
Bulgaria, Cyprus, Slovenia	The national schemes are mobile-first by design and steer citizens to a phone app.
France	Carte Vitale and CPS require homologated terminals, which is a certification we do not hold. The tachograph card is unaffected and works.
Greece	The Greek ID chip is contactless RFID, not a contact chip.

1.5 Outside Europe

1.5.1 United States and Canada — CAC and PIV

Works with the contact reader. DoD CAC and federal PIV cards are contact cards for logical access, and the contact reader drives them.

> **⚠ Warning**
>
> **A contactless reader cannot be used to log in with a PIV or CAC card.** Under NIST SP 800-73, the card's PIN cannot be verified over the plain contactless interface — only over the contact interface or a Virtual Contact Interface, and establishing a VCI requires secure-messaging key establishment plus a pairing code. Without a PIN there is no authentication. The contactless interface exposes the card's physical access control credential only. Any product advertising contactless CAC login is describing its *contact* slot.

1.5.2 United Arab Emirates — Emirates ID

Works with either reader. The Emirates identity authority's own developer documentation states that any PC/SC reader, contact or contactless, may be used. Note that the practical audience is organisational — registration desks, telecoms, banks, healthcare — because the consumer route is the UAEICP phone app.

Middleware: EIDA SDK Toolkit.

1.6 Travel documents — ICAO 9303

EU Regulation 2019/1157 requires member state ID cards to follow **ICAO Doc 9303**, the same specification as passports. The contactless reader reads ICAO 9303 electronic machine-readable travel document chips over ISO 14443 Type A and B, with extended APDU — which matters, because ePassport data groups routinely exceed the short-APDU limit and readers without extended APDU fail on them.

What that does **not** mean:

- The chip is password-protected. Access keys are derived from the **MRZ** (document number, date of birth, expiry date) or from the **CAN** printed on the document.
- It requires **MRTD reading software**, which Cryptnox does not supply.

- It yields document data and the facial image only. **Fingerprints are protected by Extended Access Control and are available to authorities only.**
- It is not the eID authentication or signature function, which is a different applet.

1.7 Limitations

- **125 kHz cards** — EM4100, HID Prox and similar low-frequency badges. The contactless reader is 13.56 MHz only.
- **iPhone and iPad** — neither reader works with them.
- **Contact and contactless at the same time** — these are two separate products.
- **PIN entry on the reader** — neither has a keypad, so schemes that mandate on-device PIN entry are out of scope.
- **Click-to-Tap with a government card** — the button answers the FIDO2 user-presence prompt for Cryptnox FIDO2 cards on Windows. It has no effect on eID, PIV or CAC sign-in.
- **Guarantee the government's software.** We can state that the reader meets a scheme's published requirements. We cannot promise how a given public-service application behaves.

1.8 Common questions

What kind of card reader do I need for the Belgian eID? A contact smart card reader with PC/SC support, plus the free eID Middleware from eid.belgium.be. Generic readers work; there is no approved-model requirement.

Which USB card readers work with the Czech eObčanka? Any reader meeting the Ministry of the Interior's published specification — ISO 7816, CCID, PC/SC, ideally WHQL-certified and plug-and-play. There is no approved-model list.

Does the Click-to-Tap button work with my national eID card? No. Click-to-Tap answers the FIDO2 user-presence prompt, and only for Cryptnox FIDO2 smart cards on Windows. National eID, PIV and CAC cards are unaffected by it: whether they work depends on their interface and their own middleware, exactly as set out above.

Why does my reader work in one country but not another? Three things have to line up: the card's interface (contact or contactless), the reader's specification (PC/SC, extended APDU, voltage class), and the national middleware. In practice the middleware is the most common blocker — it may support only one interface, be restricted to an approved list of reader names, or not be publicly downloadable at all.

Do I need extended APDU support? For the German Personalausweis, ICAO 9303 travel documents, and any scheme using 3072- or 4096-bit RSA keys, yes. Certificates and signatures at those sizes exceed the 255/256-byte short-APDU limit, and a reader without extended APDU fails on them.

Does one reader cover several countries? For contact cards, largely yes — Belgium, Czechia, Estonia, Slovakia, Hungary, Latvia, Romania, Portugal, Spain, Finland and the UK/Irish tachograph card all run on a standards-compliant PC/SC contact reader. The middleware differs per country; the hardware does not.

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1.9 Related

- *FIDO2 card compatibility — services and platforms* — which online services and operating systems work with a Cryptnox FIDO2 card, as a second factor or passwordless, and where extra software is needed.

1.10 Sources

Every verdict on this page is drawn from the scheme's own published reader requirements and, where available, from user reports of generic readers in service. Government middleware pages are linked inline above. Standards referenced: ISO/IEC 7816, ISO/IEC 14443, ICAO Doc 9303, NIST SP 800-73, EU Regulation 2019/1157.

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2 FIDO2 Card Compatibility — Services and Platforms

Last reviewed: 26 August 2026.

This page answers: **will a Cryptnox FIDO2 card sign me in to a given service, on the device I actually use?**

The answer is never a single yes or no, because FIDO2 compatibility has two independent halves. Read your **platform** row, read your **service** row, and take the intersection.

On this page

- *Table A — platform and transport*
- *Table B — services*
- *The composition rule*
- *Limitations*
- *Common questions*
- *Related*

2.1 Table A — platform and transport

This decides **which protocol is available at all**. It is a property of the operating system and the connection, not of the service.

Platform	Transport	Protocol available	Notes
Windows 10/11	NFC reader or contact reader	full FIDO2 / CTAP2	Broadest support. Works across all major browsers.
iOS	NFC (hold card to top of phone)	FIDO2 / CTAP2	iPhone 7 and later, iOS 13.3+.
Android	NFC	**CTAP1 / U2F only**	Android does **not** support CTAP2 over external NFC security keys. Most services keep U2F backward compatibility, so second-factor sign-in works — but see the worked example below.
macOS	NFC reader or contact reader	varies	Support differs by macOS version and browser. Test before relying on it.
Linux	contact reader	FIDO2 **via a bridge**	Browsers expect a HID interface; CCID smart card readers do not present one. Requires the open-source Cryptnox FIDO2 HID bridge, a small daemon that presents the card to the browser as an HID-FIDO device. **This applies to every FIDO2 use on Linux** — second factor, passwordless, everything.

2.2 Table B — services

This decides **what the service accepts**. Nearly every service that supports security keys uses them as a **second factor**; passwordless is a smaller, explicitly enabled set.

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Service	Second factor	Passwordless	Setup guide
Microsoft (Entra ID / Account)	yes	**yes** (Entra ID)	guide · overview
Google	yes	**yes** (Advanced Protection)	guide · overview
AGOV (Switzerland)	yes	**yes**	registration guide
SwissID (Switzerland)	yes	—	registration guide
login.gov (United States)	yes	**no**	Always requires a password in addition to MFA — the key is a second factor, never the only one. overview
Apple	yes	—	overview
GitLab	yes	—	guide
Facebook	yes	—	guide
X / Twitter	yes	—	guide
Cloudflare	yes	—	overview
Shopify	yes	—	overview
Binance	yes	—	overview
Coinbase	yes	—	overview
Bitfinex	yes	—	overview
Bank of America	yes	—	overview
Dropbox	yes	—	overview
NordVPN	yes	—	overview
Fastmail	yes	—	overview
GoDaddy	yes	—	overview
SSH	yes	—	guide

A dash under *Passwordless* means the service has not enabled FIDO2-only sign-in, or we have not confirmed that it has — not that it is impossible. Test before relying on it.

2.3 The composition rule

Your answer is the intersection of your platform row and your service row. Both must allow what you are trying to do.

The worked example is **Microsoft on Android**:

- the card is CTAP2-capable ✓
- Microsoft Entra ID is CTAP2-capable ✓
- **the combination still fails ✗**

Android speaks only CTAP1/U2F over an external NFC key, and Microsoft requires CTAP2 with user verification on every attempt — it does not accept U2F credentials. Use a reader on Windows, or NFC on iPhone.

That single case teaches the whole model: **capability is not the same as a usable path.** A component being able to do something says nothing about whether the platform in front of it

will ask.

Two related consequences worth stating plainly:

- **One card can appear twice in an account.** Registering the same card over CTAP1 on Android and CTAP2 on desktop creates two separate credentials. That is expected.
- **Cryptnox FIDO2 is listed by Microsoft** for Entra ID attestation (AAGUIDs `9c835346-796b-4c27-8898-d6032f515cc5` and `1d1b4e33-76a1-47fb-97a0-14b10d0933f1`, both NFC). Attestation eligibility is not the same as passwordless support, and individual server implementations of CTAP1 vary.

2.4 Limitations

- **Passwordless everywhere.** These are **MFA-first**: a hardware second factor on top of a password. Passwordless works only where a service has explicitly enabled FIDO2-only sign-in. login.gov is the clearest counter-example — it always requires a password.
- **CTAP2 over Android NFC.** Second-factor U2F works on most services; Microsoft does not.
- **Linux without the bridge.** A CCID reader alone will not satisfy a browser.
- **Fingerprint on the card.** FIDO2 user verification on these cards is **PIN-based**.
- **Migrate credentials between cards.** Each card holds its own keys. A replacement card must be registered with each service again.
- **Act as a USB device.** These are NFC and contact smart cards, not USB tokens.

2.5 Common questions

Does a FIDO2 security key card work with Android over NFC? Yes for second-factor sign-in on most services, because Android supports CTAP1/U2F over external NFC keys and most services keep U2F backward compatibility. It does not work with Microsoft, which requires CTAP2 with user verification.

Which services support passwordless sign-in with an external NFC card? A small set that has explicitly enabled FIDO2-only login — Microsoft Entra ID, Google Advanced Protection and AGOV among them. Most services accept the key only as a second factor on top of a password, and login.gov always requires one.

Why does my security key appear twice in my account settings? Because CTAP1 and CTAP2 registrations create separate credentials. Registering the same card on an Android phone and on a desktop produces two entries. Both work.

What decides whether a given service and device combination works? Three things: the card's capability, the platform's protocol support, and what the service accepts. In practice the platform is the usual blocker — most notably Android, which speaks only CTAP1 over external NFC keys.

Do I need extra software? On Linux, yes — the open-source Cryptnox FIDO2 HID bridge, for every FIDO2 use. On Windows, macOS and iOS, no. Setting or changing the card PIN uses the Cryptnox app.

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2.6 Related

- FIDO2 card technical specifications
- Using the card with Android
- Windows setup guide
- Setting the card PIN
- Managing credentials on the card
- *Smart card reader compatibility by country*

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3 Click-to-Tap: FIDO2 User Presence on a Desktop Reader

Last reviewed: 29 August 2026.

This page explains one button: what it is for, what it requires, and what it does not do.

3.1 The problem it solves

FIDO2 and WebAuthn require a **user-presence check** on every authentication. The authenticator must prove that a human, and not software running unattended in the background, approved the sign-in. It is a deliberate part of the standard, not an implementation detail.

A USB security key satisfies that check with a metal touch button. A smart card does not have one. When the card sits in an ordinary contact reader, the only gesture left that the host recognises as "a person did something" is **removing the card from the slot and inserting it again** — once per sign-in.

That is slow, it wears the card contacts and the reader connector, and the card is out of the reader at exactly the moment the user is trying to work.

3.2 How Click-to-Tap works

Pressing the button drives the smart-card interface through a controlled **extraction-and-reinsertion sequence at the protocol level**. There is no mechanical movement: the card is never physically ejected. The host sees the card leave and return, which is the event it was waiting for, and the card stays seated in the slot.

The effect is the contactless "tap" gesture, delivered on a contact-mode card.

3.3 Requirements

Requirement	Detail
Card	**Cryptnox FIDO2 smart cards only.** The button drives a Cryptnox-specific signalling sequence; it is not a generic CCID feature.
Operating system	**Windows.** The button has no effect on macOS or Linux.
Reader	The Cryptnox Click-to-Tap USB-C contact reader. The contactless reader has no button and does not apply.
Software	None for the button itself. The reader is a standard USB CCID / PC-SC device and uses the smart-card stack already present in Windows.

With any other card, or on any other operating system, the device remains a fully functional standard CCID / PC-SC contact reader — you simply do not get the tap simulation.

3.4 Using it

1. Plug the reader into a USB port. The indicator confirms it has power.
2. Insert the Cryptnox FIDO2 card into the full-size slot, chip contacts down.
3. Sign in to a FIDO2 / WebAuthn service as usual and choose the security key.
4. When the prompt asks you to **tap your security key**, press the button on the reader.
5. The sign-in completes. Leave the card in the slot for the next one.

3.5 Troubleshooting

The button does nothing. Check the three requirements above in order: it must be a Cryptnox FIDO2 card, on Windows, in the Click-to-Tap reader. Any one of the three missing produces exactly this symptom, with no error message.

The service never asks me to tap. Then there is nothing for the button to answer. Some services complete without a user-presence prompt, and some browsers prompt only on first registration. The button is only meaningful while a prompt is waiting.

The card is not detected at all. That is a reader or middleware question, not a Click-to-Tap question. Confirm the card is seated with the contacts facing down and fully inserted. On Linux, install `pcscd`, `libccid` and `pcsc-tools`, then check that `pcsc_scan` sees the reader.

FIDO2 sign-in does not work on Linux. Browsers on Linux expect an HID FIDO interface, which a CCID reader does not present by itself. Install the open-source Cryptnox FIDO2 HID bridge. Note that this enables FIDO2 sign-in on Linux — it does not enable the Click-to-Tap button, which remains Windows-only.

I want to use the button with my eID, PIV or CAC card. Not possible, and not a defect. Click-to-Tap answers a FIDO2 prompt. Government cards authenticate through their own middleware and are unaffected by it. See *Smart Card Reader Compatibility by Country* for what those cards need.

3.6 What Click-to-Tap is not

- **Not a replacement for the PIN.** User presence and user verification are different checks. The card's PIN policy is unchanged.
- **Not a security boundary.** The button is a convenience for a gesture the standard requires; it neither adds nor removes cryptographic protection.
- **Not a secure element.** The reader holds no keys, stores nothing and signs nothing. Every cryptographic operation happens on the card.
- **Not applicable to contactless cards**, which already tap physically.

3.7 Provenance

The Click-to-Tap virtual-button signalling is Cryptnox-engineered firmware layered on the standard USB CCID interface; the underlying reader hardware is a commodity component. The button is protected by registered Austrian utility model **GM 55114/2025**, registered German utility model **No. 202025108028.1**, and United States patent application **No. 19/534.472** (pending).

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